Math 3680 — Final Project

A Typeset Document of Notes from CSC 3665 —
Introduction to Database Management Systems

Arjan Singh Ellingson

October 19, 2026

Table of Contents

1 Basic Relation Algebra 2
1.1 Relational Data Model . . . 2
1.2 Relational Algebra . . . 2

2 ER Diagrams 3

3 Normalization 3

4 Basic SQL Statements 4

5 Advanced Relational Algebra 4

6 From English To SQL to Rel. Alg. 4

7 External Content Referenced 4

Index of Terms 5

1 Basic Relation Algebra

The class started with very basic relational algebra, expanding on set theory and cardinality discussed in CSC 248 (now MATH 2031).

1.1 Relational Data Model

- *Tuple*: A list of grouped-together attributes. Synonymous to row/record.
- *Attributes*: A list of columns, all values that a single fields can be.
- *Relation*: A set of data organized as tuples. The entire table is the relation.
- *Key*: an attribute (column) that can uniquely identify rows so that relationships can be established between tables. Selection process goes (generally)

$$\begin{aligned}\text{super keys} &\supseteq \text{candidate keys} \\ &\supseteq (\text{one chosen as}) \supseteq \text{primary key} \\ &\supseteq (\text{candidate keys} - \text{primary key}) = \text{alternate key}\end{aligned}$$

- *Super Key*: Attributes that can uniquely identify tuples. May include extraneous values.
- *Candidate Key*: A minimal super key that can identify a row uniquely. There may be many; rows that are not chosen are called *alternate keys*.
- *Primary Key*: a chosen unique non-null candidate key to identify rows.
- *Foreign Key*: A shared value / column between two tables that allows linking of tables. For example,

$$S_1 = \{StudentID,\ Dept,\ Credits\}$$

$$S_2 = \{Dept_ID, Dept_Name\}$$

- *Composite Key*: A set of 2+ attributes that can together uniquely identify a tuple.
- *Natural Key*: A key coded with real world meaning, such as an email
- *Surrogate Key*: A key with no real world value. Created only for identification purpose, often a UUID or hex code.

1.2 Relational Algebra

Given two schemas R and S

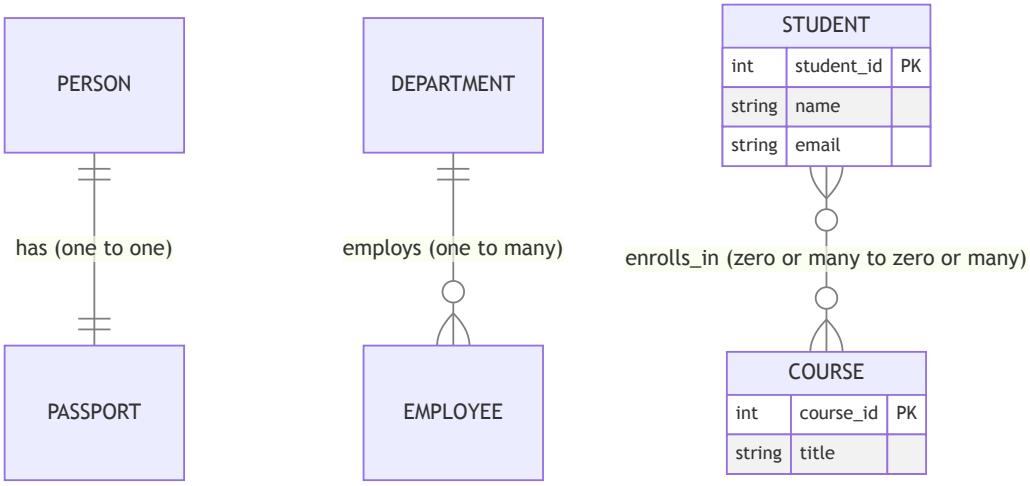
- $R \cup S$: What will the table look like with data from both? Note, schemas must match, there will be more columns.
- $R \cap S$: What tuples are in both? Note, schemas must match.
- $R - S$: What is in R but not S?
- $R \times S$: Every possible tuple of R paired with every tuple of S
- $\sigma_c(R)$: Show only rows of R satisfying condition c
- $\pi_{(A)}(R)$: Picks columns A and removes any duplicate values.
- $R \bowtie S$: adds more columns based on at lease one shared attribute. This makes the schema larger.
- $R \bowtie_\theta S = \sigma_\theta(R \times S)$ tuples that satisfy θ.

2 ER Diagrams

This section taught the basics of *entity-relationship diagrams* () and how to use them to model a database.

Not only did we learn how to diagram, but also Lab 2 emphasized coding our own, which was done in the programming language *mermaid*, which fits directly into LaTeX.

Below is a simple example of 3 different types of relationships, one-to-one, one-to-many, and many-to-many.



Note that the symbols next to a node A represent how many of A can be in B. For example, a department has many or zero employees, but an employee must have exactly one department.

3 Normalization

Take for example the following table:

StudentID	**Email**	**SSN**	FirstName	LastName	EnrollmentDate
1001	alice@univ.edu	123-45-6789	Alice	Smith	2024-01-15
1002	bob@univ.edu	987-65-4321	Bob	Johnson	2024-01-15
1003	carol@univ.edu	555-55-5555	Carol	Williams	2024-02-01

Table 1: STUDENT table demonstrating key types.

4 Basic SQL Statements

5 Advanced Relational Algebra

6 From English To SQL to Rel. Alg.

7 External Content Referenced

Here is where I wanted to link any external resources I used to help in learning the material outside of Cal Poly IP.

Index of Terms

alternate keys, *see* Candidate Key, 2
Attributes, 2

Candidate Key, 2
Composite Key, 2

entity-relationship diagrams, 3
ER Diagrams, *see* Entity-Relationship Diagrams

Foreign Key, 2

Key, 2

mermaid, 3

Natural Key, 2

Primary Key, 2

Relation, 2

Super Key, 2
Surrogate Key, 2

Tuple, 2